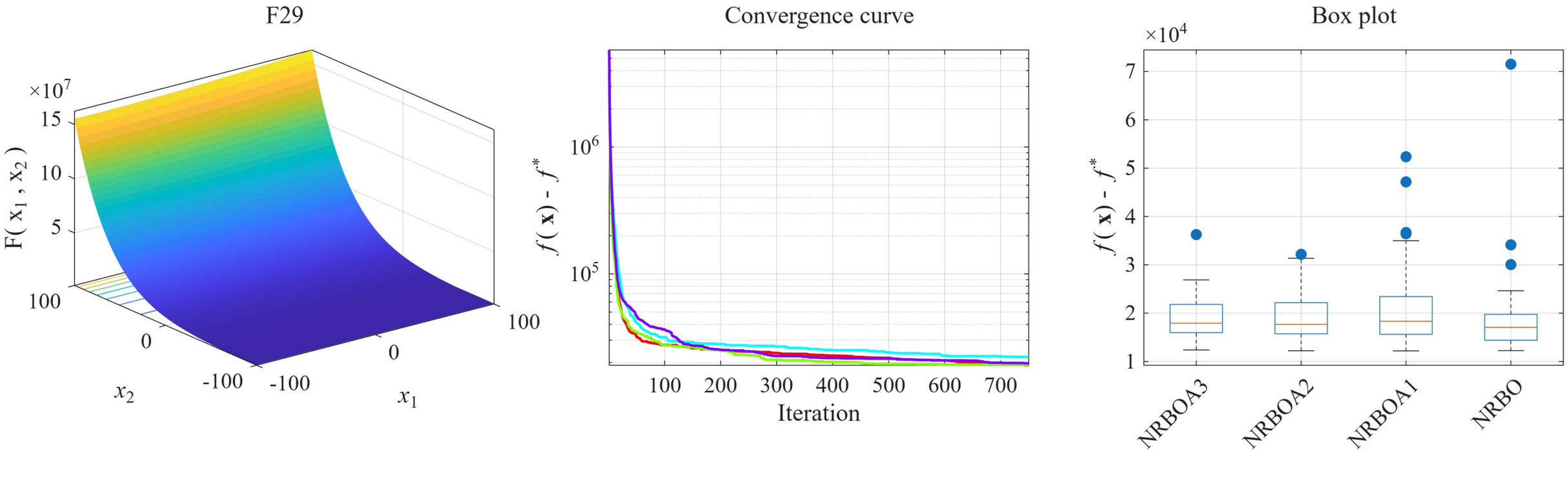

F29
×10^7
15
10
5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^6
10^5
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
7
6
5
4
3
2
1
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO